Fields and Waves I

Lecture 2

Sine Waves on Transmission Lines I

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Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Transmission Line History
- Transmission Line Model
- Telegrapher's Equations: Time Domain
- Traveling Waves
- Telegrapher's Equations: Phasor Domain



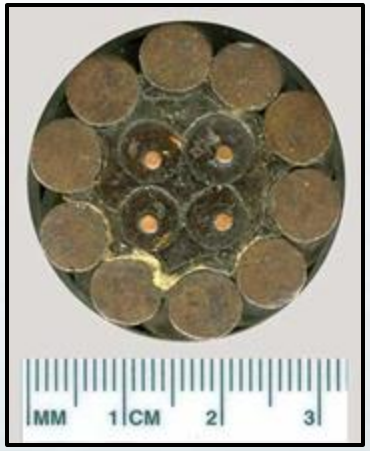
Transmission Line History Review

- One of the first practical applications of circuit theory was the telegraph (1816)
- Long telegraph cables suffered from problems:
  - Voltage drop due to resistance
  - Signal delay
  - Signal distortion
- The equations that made up this model became known as the Telegrapher's Equations
- These equations allowed for intercontinental telegraph lines to develop, and they are used for many modern applications such as CPU microstrip lines

Transmission Line Model

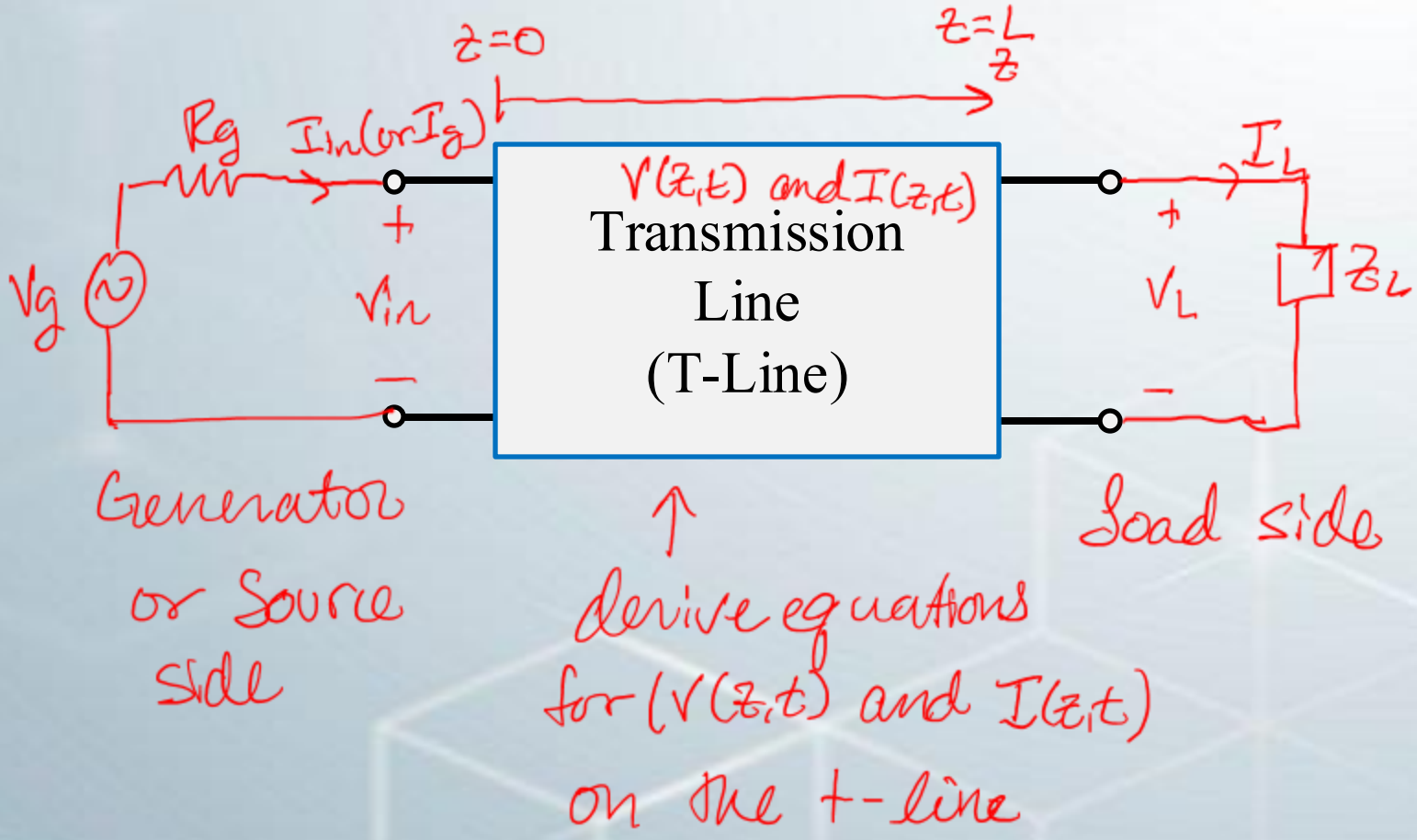
Transmission Line Model

- The purpose of a transmission line is to **transmit signal and/or power from one place to another.**
- Examples include:
  - Power Lines (60Hz)
  - Coaxial Cables
  - Twisted Pairs
  - Interconnects inside a computer



- Note that these all have **two conductors.**
- In transmission line theory, we represent a transmission line as having certain properties **per unit length**.

Transmission Line Model (2-Port Network)

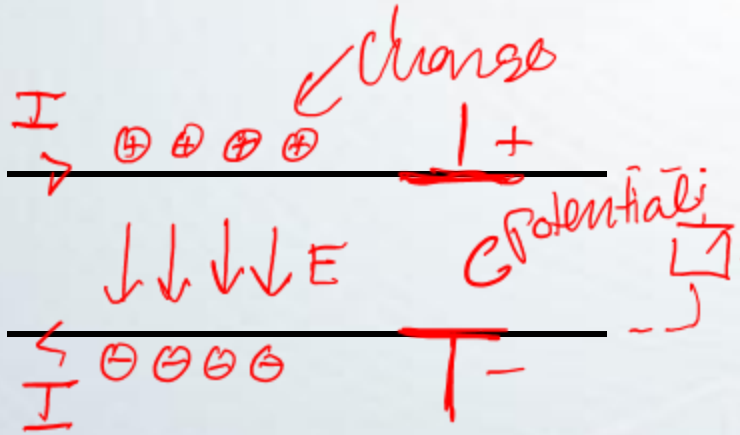


Transmission Line Model

Cables have both ***L*** and ***C***:

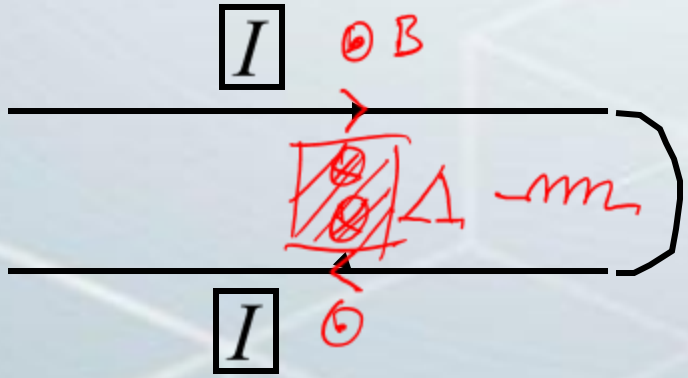
2 wire example:

C



C between conductors

L



L is a series effect



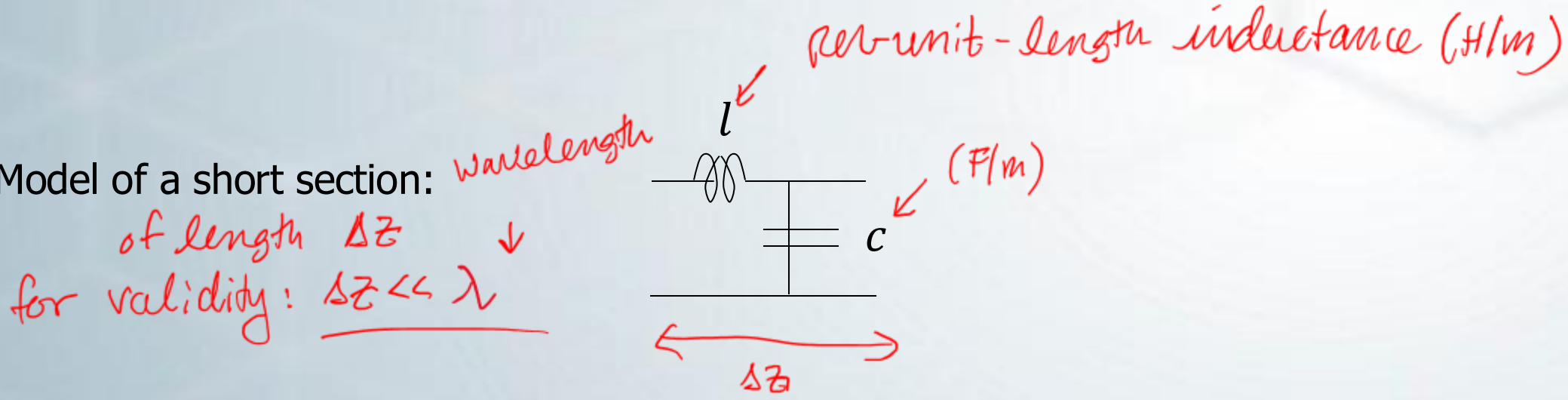
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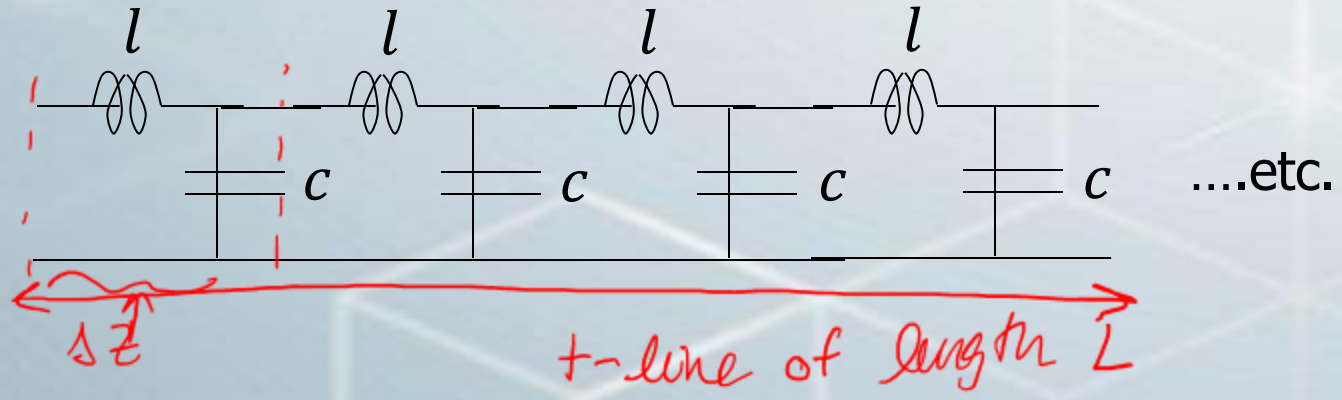
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Transmission Line Model

Model of a short section:



Model the full length as:



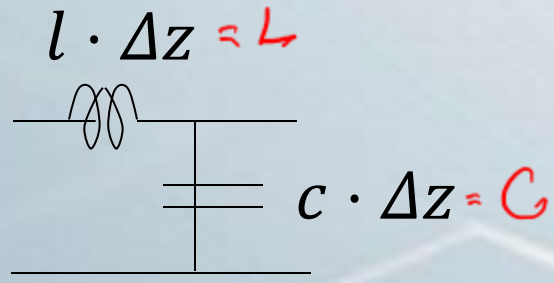
....etc.

Transmission Line Model

Since ***L*** and ***C*** are **distributed through the length of the cable** rather than discrete lumps of inductance/capacitance,

l and c are defined per unit length of cable

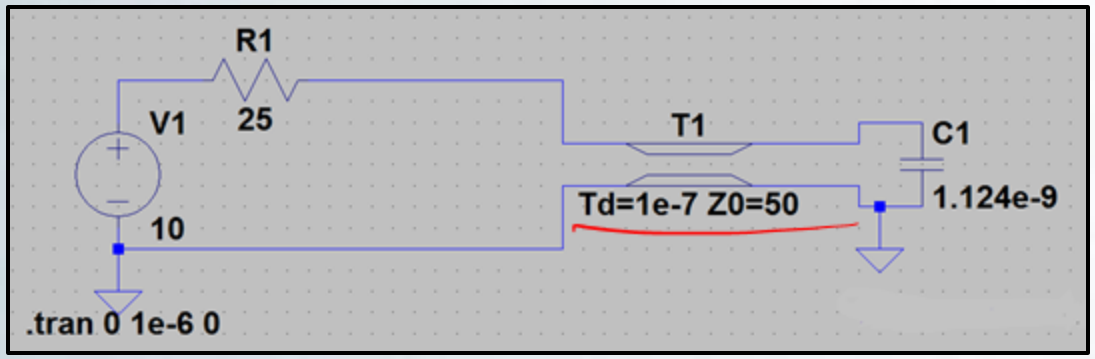
If **Δz** represents a length of cable:



l = inductance/length

c = capacitance/length

Transmission Line Model



- If transmission lines have no losses → treat them as a "black box" with just two quantities: the **time delay**, and the **characteristic impedance.**
- **Time delay**: how long a signal takes to propagate through the line. (Keep in mind that for higher frequency AC signals, a small time delay can mean a big phase change).
- **Characteristic impedance**: the voltage-to-current ratio for a signal on the line, *or,* what the impedance of the line "looks like" to a signal entering the line when there is no signal on it

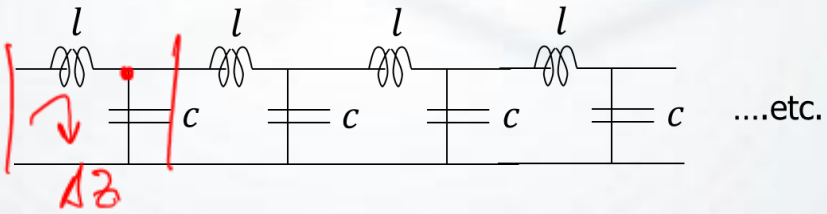
Transmission Line Model

- This simple "black box" analysis is good enough for some purposes, such as calculating how long a signal will take to reach its destination.

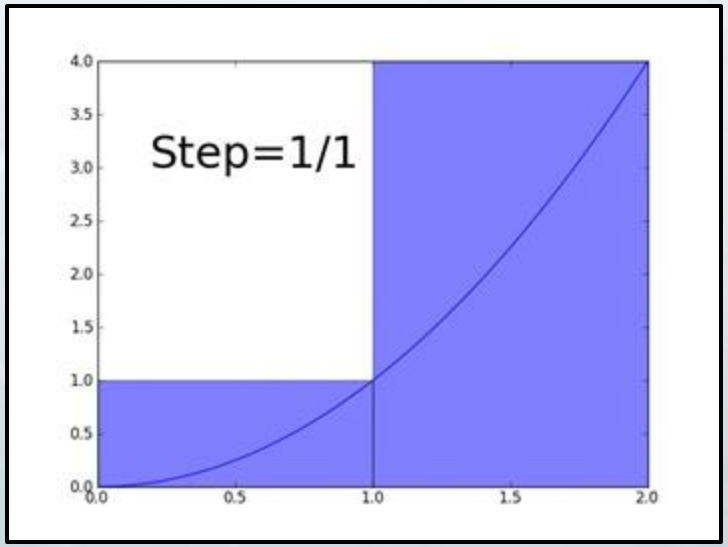
- **Complete analysis** of transmission lines means being able to calculate the voltage and current anywhere on the line at any time. We do this using the **Telegrapher's Equations**.

Transmission Line Model

- Heaviside et. al. derived the telegrapher's equations by thinking of the lumped-element transmission line and letting the length that each element represented approach zero.



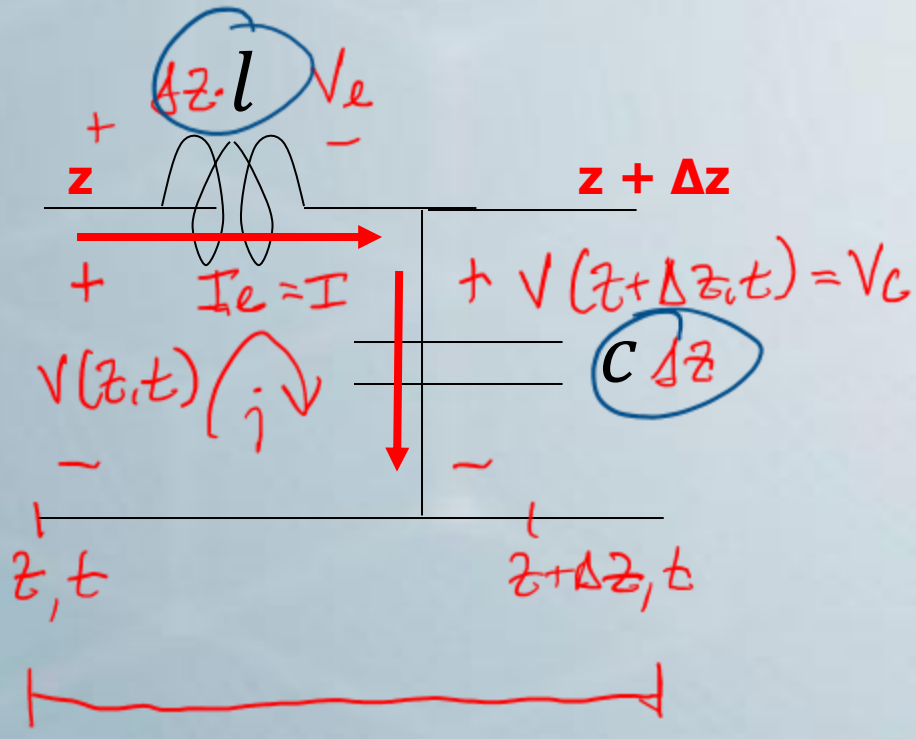
- This is a bit like the rectangle area approximation method of integrals in calculus.



Telegrapher’s Equations: Time Domain

Telegrapher's Equations

$$V_\ell = L\frac{dI_\ell}{dt} = \Delta z \cdot \ell \frac{dI_\ell}{dt}$$



total L of $L = \ell \cdot \Delta z$

C of $C = c \cdot \Delta z$

Applying Kirchhoff's voltage law:

$$-V(z,t) + V_\ell + V(z+\Delta z,t) = 0$$

$$-V(z,t) + \Delta z \cdot \ell \cdot \frac{dI_\ell(z,t)}{dt} + V(z+\Delta z,t) = 0$$

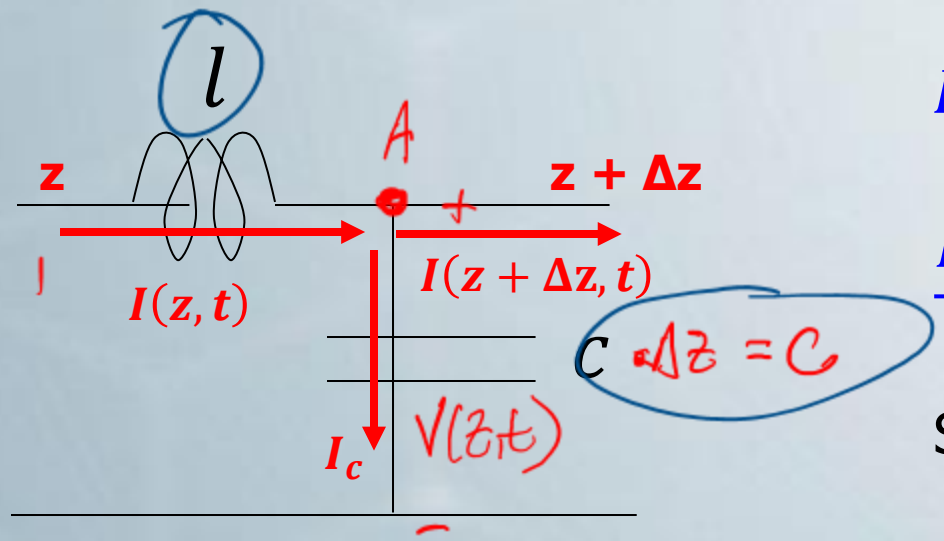
Rearranging:

$$\frac{V(z+\Delta z,t) - V(z,t)}{\Delta z} = -\ell \frac{dI(z,t)}{dt}$$

Letting $\Delta z \to 0$:

$$\frac{\partial V(z,t)}{\partial z} = -\ell \frac{\partial I(z,t)}{\partial t}$$

Telegrapher's Equations



$$I_c = C\frac{dV_c}{dt}$$

Applying Kirchhoff's current law:

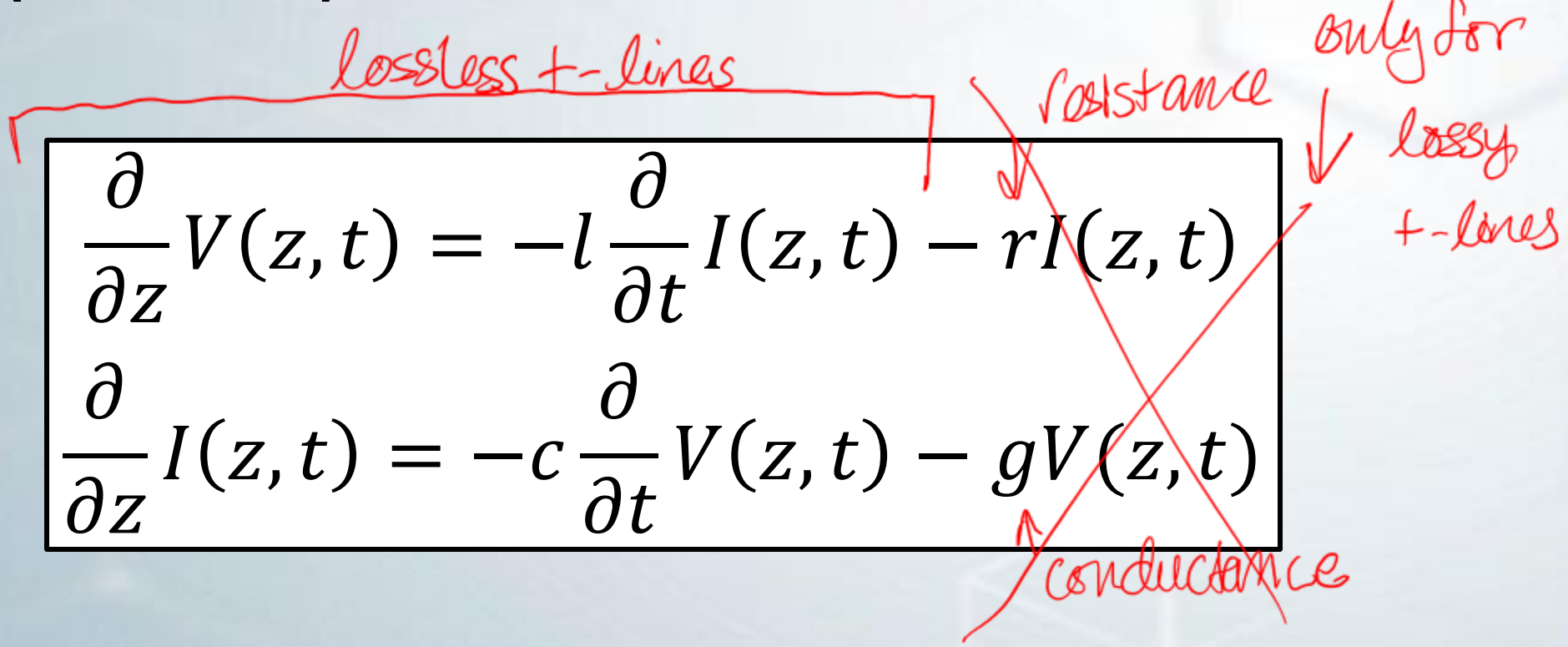
$$I(z,t) - c\Delta z\frac{\partial V_c}{\partial t} - I(z+\Delta z,t) = 0$$

$$\frac{I(z+\Delta z,t) - I(z,t)}{\Delta z} = -c\frac{\partial V_c(z,t)}{\partial t}$$

Similarly letting Δz → 0:

$$\frac{\partial I(z,t)}{\partial z} = -c\frac{\partial V_c(z,t)}{\partial t}$$

Telegrapher's Equations



$$\frac{\partial}{\partial z}V(z,t) = -l\frac{\partial}{\partial t}I(z,t) - rI(z,t)$$

$$\frac{\partial}{\partial z}I(z,t) = -c\frac{\partial}{\partial t}V(z,t) - gV(z,t)$$

These equations are clearly coupled… so how do we actually find V(z,t) and I(z,t)?

Telegrapher's Equations

To find the solution, combine the two equations:

Differentiate (1) with respect to z:

$$\frac{\partial^2}{\partial z^2}V(z,t) = -l\frac{\partial}{\partial z}\left(\frac{\partial}{\partial t}I(z,t)\right)$$

Plug in (2):

$$\frac{\partial^2}{\partial z^2}I(z,t) = -c\frac{\partial}{\partial z}\left(\frac{\partial}{\partial t}V(z,t)\right)$$

We obtain the following PDE:

$$\frac{\partial^2}{\partial z^2}V(z,t) = lc\frac{\partial^2}{\partial t^2}V(z,t)$$

This is a wave equation!

general solutions are sinusoids (or complex exponentials)

Traveling Waves

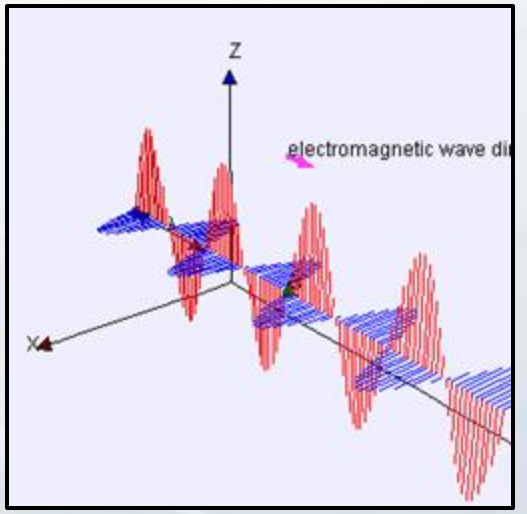
Traveling Waves

The "Waves" Part of Fields & Waves

Electromagnetic fields propagate as waves. They are governed by the electromagnetic wave equations, which are derived from Maxwell's Equations and can be written as:

$$\frac{\partial^2 \vec{E}}{\partial t^2} = c^2 \nabla^2 \vec{E} \qquad \frac{\partial^2 \vec{B}}{\partial t^2} = c^2 \nabla^2 \vec{B}$$

where **c** is the speed of light / EM waves. (Note that c depends on the medium the wave is propagating through!)



Source: https://en.wikipedia.org/wiki/Electromagnetic_radiation

Traveling Waves

- Rearranging our equation:

$$\frac{\partial^2 V}{\partial z^2} = lc\frac{\partial^2 V}{\partial t^2} \rightarrow \frac{\partial^2 V}{\partial t^2} = \frac{1}{lc}\frac{\partial^2 V}{\partial z^2} \rightarrow \frac{\partial^2 V}{\partial t^2} = u^2\frac{\partial^2 V}{\partial z^2}$$

standard form

In this standard form of the wave equation, u represents velocity. So for our transmission line voltage waves,

$$u = \frac{1}{\sqrt{lc}}$$

generally given in terms of a "velocity factor"

$0 < Vf < 1$

where $u = Vf \cdot c$ ← speed of light in vacuum

Traveling Waves

Sinusoids represent a solution to these equations:

$$v(z,t) = A\cos\left(\omega t \mp \frac{\omega}{u} z\right) = A\cos(\omega t \mp \beta z)$$

angular freq. (rad/s)

$$\omega = 2\pi f$$

freq. in Hz

$$\beta = \frac{\omega}{u} \quad \frac{\text{rad/s}}{\text{m/s}} = \text{rad/m}$$

phase constant

(change in phase per unit length) = $\dfrac{\text{angular velocity (change in phase per unit time)}}{\text{wave velocity (unit length per unit time)}}$ = $\dfrac{\Delta \text{phase}}{\Delta \text{space}}$

Traveling Waves

Sinusoids represent a solution to these equations:

$$v(z,t) = A\cos\left(\omega t \mp \frac{\omega}{u} z\right) = A\cos(\omega t \mp \beta z)$$

$$\omega = 2\pi f \geq 0 \qquad \beta = \frac{\omega}{u} \geq 0$$

Also note that the **sign in front of the phase constant** (and therefore the wave velocity) can be either positive or negative. This represents the fact that the wave can be traveling either forward or backward.

- If the phase constant has a $+$ sign in front of it, the wave is traveling in the "-z" direction
- If the phase constant has a $-$ sign in front of it, the wave is traveling in the "+z" direction

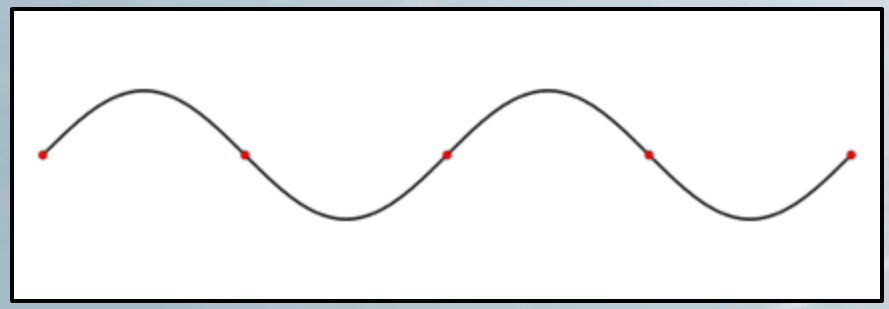
Traveling Waves



Traveling wave (depends on t and z in the argument of one sinusoid)

$$A\cos(\omega t \mp \boldsymbol{\beta} z)$$

in other words you can relate t and z with u.



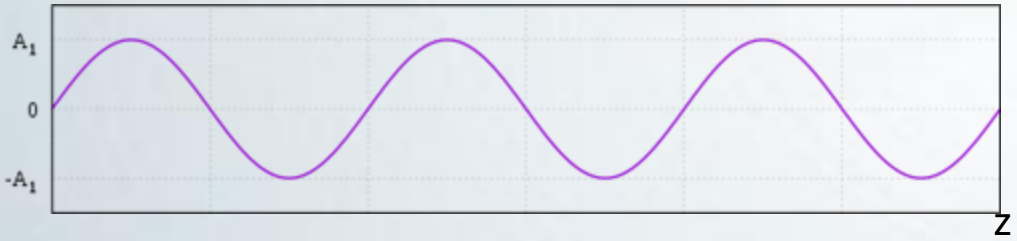
Standing wave (depends on t and z but as two different terms)

$$A\sin(\omega t)\cos(\boldsymbol{\beta} z)$$

No relationship between t and z.

Traveling Waves

$$A\cos(\boldsymbol{\beta} z)$$



A static sine wave varies in space but not time.

What would A sin(ωt) look like?

Does not vary in space: value that increases and decreases sinusoidally everywhere at the same time – not a wave in the physical sense

Traveling Waves

$$\cos\left(\omega t - \frac{\omega}{u} z\right) = \cos(2\pi f t - \frac{2\pi f}{u} z)$$

- Consider one other property. What is the distance required to change the phase of this expression by 2π?

λ : wavelength

- In other words, this is an expression for wavelength.

Traveling Waves

Some Properties Summarized

Solutions look like $A\cos(\omega t \mp \beta z)$

Phase constant

$$\beta = \frac{\omega}{u} = \omega\sqrt{lc} = \frac{2\pi}{\lambda}$$

Frequency

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Wavelength

$$\lambda = \frac{2\pi}{\beta} = \frac{u}{f}$$

Velocity

$$u = \frac{1}{\sqrt{lc}}$$

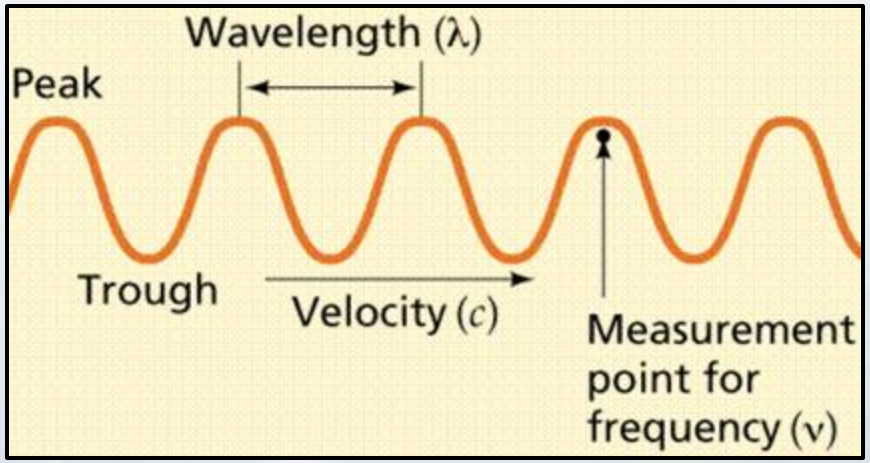


Figure from http://www.emc.maricopa.edu/

Traveling Waves

Do Lecture 02 Exercise on Gradescope

Telegrapher’s Equations: Phasor Domain

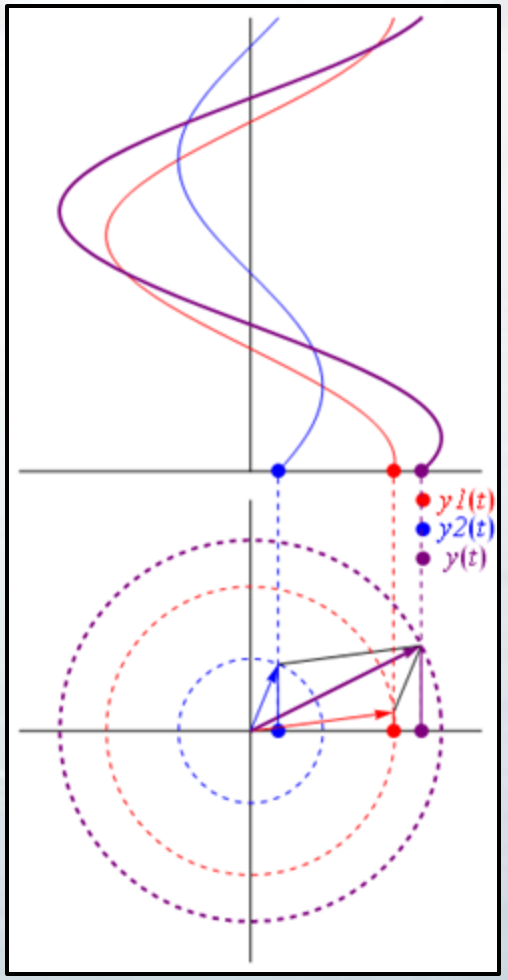
Phasor Notation

- What are phasors?
- In short, they are the **periodically time-variant version of vectors.**
- Working in phasor domain temporarily collapses the periodic time-variant component, allowing a linear expression to be written.
- This is useful for solving current or voltage expressions in linear circuits (see Ulaby pg. 38-41… see ECSE 2010)
- Especially useful in power engineering, where electric grids can contain *many* different time-varying voltages/currents

$$A\cos\left(\omega t+\beta x+\phi_0\right)\Leftrightarrow \tilde{A}e^{j\left(\beta x+\phi_0\right)}$$

Phasor Notation

- At right is a visualization of phasor addition.
- Working in phasor domain allows you to add these two sinusoidal functions together easily without dealing with sine and cosine addition formulas, etc.
- Many useful properties can be derived for converting between phasor and time domain (see Ulaby pg. 40)



https://en.wikipedia.org/wiki/File:Sumafasores.gif

Phasor Notation

- To convert to phasor form from space-time form, phase shift to obtain a cosine, drop the time-component of the phase, then write the remaining phase in a complex exponential

$$f(z,t) = A\cos(\omega t \mp \beta z + \theta) = Ae^{j(\mp\beta z+\theta)}$$

- If A is complex,

$$A = |A|e^{j\phi}$$

$$f(z,t) = A\cos(\omega t \mp \beta z + \theta) = |A|e^{j(\mp\beta z+\theta+\phi)}$$

Phasor Notation

- To convert to space-time form from the phasor form, multiply by $e^{j\omega t}$ and take the real part.

$$f(z,t) = \mathrm{Re}\left[\tilde{A} e^{\mp j\beta z} e^{j\omega t}\right] = A\cos(\omega t \mp \beta z)$$

- If A is complex,

$$\tilde{A} = |\tilde{A}| e^{j\theta}$$

$$f(z,t) = \mathrm{Re}\left[|\tilde{A}| e^{j\theta_A} e^{\mp j\beta z} e^{j\omega t}\right] = |\tilde{A}|\cos(\omega t \mp \beta z + \theta_A)$$

Phasor Notation

Example

A phasor voltage is given by: $\tilde{V} = j5 = 5e^{j\pi/2}$

- How do we find V(z,t)?

$$V(z,t) = \mathrm{Re}\{\tilde{V}e^{j\omega t}\} = \mathrm{Re}\{j5e^{j\omega t}\} = \mathrm{Re}\{5e^{j(\omega t+\pi/2)}\}$$

$$= \mathrm{Re}\{5[\cos(\omega t+\pi/2) + j\sin(\omega t+\pi/2)]\} = \underline{5\cos(\omega t+\pi/2)}$$

$$= -5\sin(\omega t+\pi/2)$$

Telegrapher's Equations: Phasor Notation

- In phasor form, the 1st order differential equations from KVL and KCL are:

$$\frac{\partial i}{\partial z} = -c\frac{\partial v}{\partial t} \qquad \frac{\partial v}{\partial z} = -l\frac{\partial i}{\partial t}$$

- We have the following phasor property:

$$\frac{d}{dt}[V(z,t)] \leftrightarrow j\omega\tilde{V}$$

$\frac{d}{dt}[V(z,t)] = \frac{d}{dt}[\mathrm{Re}\{\tilde{V}e^{j\omega t}\}] \Rightarrow j\omega\tilde{V}$

- Therefore:

$$-\frac{d\tilde{V}(z)}{dz} = j\omega l\tilde{I}(z) \qquad -\frac{d\tilde{I}(z)}{dz} = j\omega c\tilde{V}(z)$$

Telegrapher's Equations: Phasor Notation

- Combining the two equations, we get two wave equations:

$$\frac{\partial^2 \tilde{V}(z)}{\partial z^2} - (j\omega l)(j\omega c)\tilde{V}(z) = 0$$

$$\frac{\partial^2 \tilde{I}(z)}{\partial z^2} - (j\omega l)(j\omega c)\tilde{I}(z) = 0$$

- We make a substitution:

$$u^2 = (j\omega l)(j\omega c) \quad \Rightarrow \quad u = j\omega\sqrt{lc} = j\beta$$

Telegrapher's Equations: Phasor Notation

- General solutions of these wave equations (from differential equations):

$$\tilde{V}(z) = V_o^+ e^{-uz} + V_o^- e^{uz}$$

$$\tilde{I}(z) = I_o^+ e^{-uz} + I_o^- e^{uz}$$

- Combining these with our earlier equations...

$$-\frac{d\tilde{V}(z)}{dz} = j\omega l \tilde{I}(z) \qquad -\frac{d\tilde{I}(z)}{dz} = j\omega c \tilde{V}(z)$$

... we can now derive:

$$\tilde{I}(z) = \frac{u}{j\omega l}\left(V_o^+ e^{-uz} - V_o^- e^{uz}\right)$$

Telegrapher's Equations: Phasor Notation

- Our two phasor expressions are now related....

$$\tilde{V}(z) = V_o^+ e^{-uz} + V_o^- e^{uz}$$

$$\tilde{I}(z) = \frac{u}{j\omega l}\left(V_o^+ e^{-uz} - V_o^- e^{uz}\right)$$

- … such that the voltage/current ratio for a given traveling wave is a constant. (In other words, we have an expression for **impedance**!)

$$Z_0 = \frac{V_0^+}{I_0^+} = \frac{-V_0^-}{I_0^-} = \frac{j\omega l}{u} = \frac{j\omega l}{\sqrt{(j\omega l)(j\omega c)}} = \sqrt{\frac{l}{c}}$$

Telegrapher's Equations: Phasor Notation

- As we discussed earlier, one high-level property of transmission lines is their **characteristic impedance, Z_0.**

$$Z_0 = \sqrt{\frac{l}{c}}$$

- Characteristic impedance can be described in a few ways:
  - The impedance that a signal "sees" at the input of transmission line when the load is far away (no signal on it)
  - The ratio of voltage and current amplitudes for a signal propagating on the line (think Ohm's Law)

Wrap-Up

- Studio session #1 on Wednesday. Bring your laptops with LTspice installed!
- My office hours: Tuesdays, 4pm – 6pm